

- 8 (a) Approximately 2×10^9 people in the world are currently infected with the bacterial disease tuberculosis (TB) caused by *Mycobacterium tuberculosis*. Early diagnosis is important so that treatment can begin.

APOPO is a non-profit organisation that has trained African giant pouched rats, *Cricetomys gambianus*, to use their sense of smell to detect *M. tuberculosis*. They do this by sniffing a sample of thick mucus from the lungs of people who may have TB. The African giant pouched rats are able to detect the presence of *M. tuberculosis* with an accuracy of 87–93%.

Fig. 8.1 shows an African giant pouched rat.



Fig. 8.1

- (i) The type of receptor cell used by African giant pouched rats to detect *M. tuberculosis* is the same as that used in human taste buds.

Name this type of receptor cell.

..... [1]

- (ii) Suggest why African giant pouched rats trained to detect *M. tuberculosis* may also be able to detect other species of *Mycobacterium* that cause TB.

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..... [2]

(b) The African giant pouched rat belongs to the kingdom Animalia in the domain Eukarya.

Complete Table 8.1 to show the full classification of the African giant pouched rat.

Table 8.1

kingdom	Animalia
.....	Chordata
class	Mammalia
.....	Rodentia
family	Nesomyidae
.....	
species	<i>gambianus</i>

[2]

- (c) Differences between members of the domain Eukarya and members of the domain Bacteria include the presence or absence of particular membrane-bound cell structures.

Outline **other** differences in the characteristic features of members of the domain Eukarya and members of the domain Bacteria.

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..... [4]

- (d) Describe, with reference to the structure of viruses, how viruses are classified.

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[Total: 11]